

Nets of 3D shapes



Key idea

A net is a flat 2D shape that folds to form a 3D shape. Different 3D shapes have different nets. Some shapes, like cylinders, have curved surfaces that flatten to rectangles.

Steps for identifying a net

1. Identify the 3D shape the net represents.
2. Count the faces and check they match the shape.
3. Visualise folding along the edges.
4. Check that opposite faces are in the right positions.

Common mistake

Thinking a cylinder has no net. A cylinder has a curved surface that unfolds to a rectangle, plus two circles.

Try these

1. How many faces does a cylinder have?
2. Name the shapes that make up a cylinder's net.
3. A cylinder has height 6 cm and radius 2 cm. What is the width of the rectangle in its net?

Cylinder net

A cylinder net has three parts:

- ▶ Two circles (top and bottom faces)
- ▶ One rectangle (curved surface unfolded)

The width of the rectangle equals the circumference of the circle ($2\pi r$). The height of the rectangle is the height of the cylinder.

